

IF330 Pemrograman Web.

03 - PHP Forms & Cookies

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PROGRAM STUDI INFORMATIKA
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REVIEW



- Data Types in PHP
- Strings
- Arrays
- Constants
- Conditional Statement
- Loop Statement
- Function

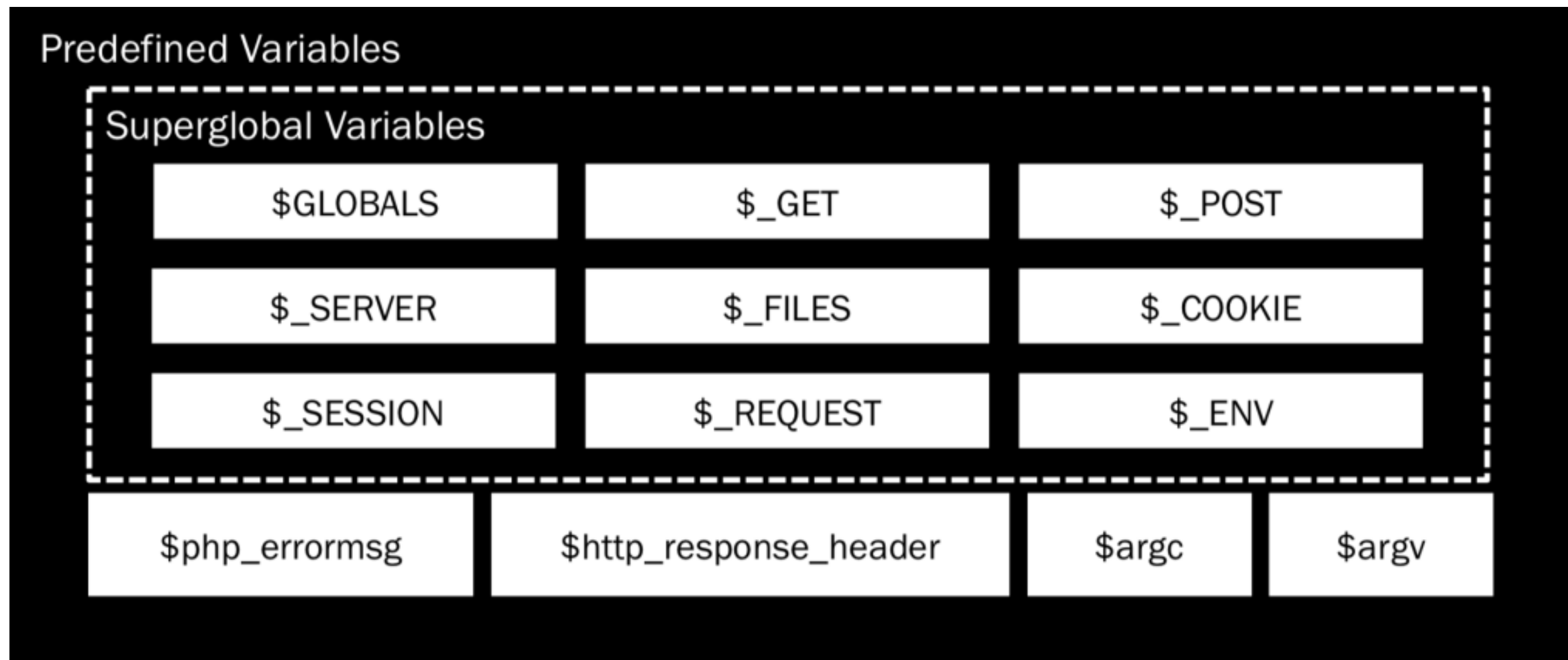
TODAY's OUTLINE



- PHP Predefined Variables
- PHP Predefined Functions
- PHP Form Submission
- HTTP Cookies
- `json_encode` & `json_decode`

PHP Predefined Variables

PHP provides a large number of predefined variables to all scripts. The variables represent everything from external variables to built-in environment variables, last error messages to last retrieved headers



PHP Predefined Variables

\$GLOBALS: References all variables available in global scope

```
<?php
function test() {
    $foo = "local variable";
    echo '$foo in global scope: ' . $GLOBALS["foo"]; // Example content
    echo '<br />$foo in current scope: ' . $foo; // local variable
}

$foo = "Example content";
test();
```

\$_SERVER: Provides server and execution environment information

```
<?php
echo $_SERVER['SERVER_NAME']; // localhost
echo $_SERVER['SERVER_PROTOCOL']; // HTTP/1.1
echo $_SERVER['SERVER_SOFTWARE']; // PHP 8.1.2 Development Server
echo $_SERVER['REQUEST_METHOD']; // GET
```


PHP Predefined Variables

\$_GET: An associative array of variables passed to the current script via the URL parameters

```
<?php
// http://localhost/index.php?name=John+Thor
echo $_GET['name']; // John Thor
```

\$_POST: An associative array of variables passed to the current script via the HTTP POST method

```
<?php
// <form action="process.php" method="post">
// <input type="text" name="prodi" value="Informatika" />
// </form>
// process.php
echo $_POST['prodi']; // Informatika
```

PHP Predefined Variables

\$_REQUEST: An associative array that by default contains the content of \$_GET, \$_POST and \$_COOKIE

```
<?php
// http://localhost/?name=John+Thor
echo $_REQUEST['name']; // John Thor
```

\$_SESSION: An associative array containing session variables available to the current script

```
<?php
session_start(); // session_start() is required
$_SESSION['course'] = "PemWeb"; // New session variable
$_SESSION['course'] = "Web Programming"; // Update session variable
```


PHP Predefined Functions

String Functions

strlen: Returns the length of given string

substr: Returns the portion of string specified by the start and length parameters

implode: Join array elements with a glue string

explode: Returns an array of substring formed by splitting a string formed by the string delimiter

```
<?php
echo strlen( string: "Hello World"); // 11
echo substr( string: "Hello World", offset: 3, length: 4); // "lo W"
$arr = array("Hello", "World");
echo implode( separator: "...", $arr); // "Hello...World"
$arr2 = explode( separator: " ", string: "Hello World");
echo $arr[1]; // "World"
```


PHP Predefined Functions

Array Functions

array: Creates an array

count: Counts all elements in an array, or something in an object

sort: Sort an array in ascending order (lowest to highest)

shuffle: Shuffle (randomize the order of the elements in) an array

```
<?php
$students = array("John Thor", "John Wick", "John Doe");
echo "<pre>";
echo $students[1] . "\n";
echo count($students) . "\n";
sort($students);
var_dump($students);
echo "\n";
shuffle($students);
var_dump($students);
```

PHP Predefined Functions

Variable Handling Functions

isset: Determine if a variable is set and is not NULL

unset: Destroys the specified variables

var_dump: Displays structured information of expressions that includes its type and value

is_XXX: Finds whether the type of variable is xxx

```
<?php
echo "<pre>";
$name = "John Thor";
if(isset($name)){
    echo '$name is set.';
}else{
    echo '$name is not set';
}

unset($name);
$data = array("John", 23, true, array(1, 2, 3));
var_dump($data);

echo is_array($data[3]);
```

Form Submission

- PHP superglobals `$_GET` and `$_POST` can be used to collect form-data
- **`$_GET`**
 - Passed via the URL parameters
 - 2000 characters limit
 - Visible to everyone (variable names and values are displayed in the URL)
 - NOT for sending passwords or other sensitive information!
- **`$_POST`**
 - Passed via HTTP POST method
 - No data amount limitation
 - Variable names and values are not visible by users
 - Supports multi-part binary input to upload files

Form Submission

form_input.php x

```
1 <form action="form_process.php" method="post">
2     <input type="text" name="data" />
3     <input type="submit" name="submit" />
4 </form>
```

form_process.php x

```
1 <?php
2
3 if(isset($_POST['submit'])){
4     echo "Data received: " . $_POST['data'];
5 }else{
6     echo "No data received.";
7 }
```

PHP Cookie

Syntax:

setcookie(name, value, expire, path, domain, secure, httponly);

Only the *name* parameter is required. All other parameters are optional.

```
<?php
$cookie_name = "user";
$cookie_value = "John Doe";
setcookie($cookie_name, $cookie_value, time() + (86400 * 30), "/");
?>

<html>
<body>
<?php
if(!isset($_COOKIE[$cookie_name])) {
    echo "Cookie named '" . $cookie_name . "' is not set!";
} else {
    echo "Cookie '" . $cookie_name . "' is set!<br>";
    echo "Value is: " . $_COOKIE[$cookie_name];
}
?>
</body>
</html>
```


json_encode()

```
<?php
echo "<pre>";
$data = array(
    "name" => "John Thor",
    "prodi" => "Informatika",
    "email" => "john.thor@umn.ac.id"
);

$data_json = json_encode($data);
var_dump($data_json);

echo "\n\n";
```

```
$data2 = array(
    "students" => array(
        array(
            "name" => "John Thor",
            "prodi" => "Informatika"
        ),
        array(
            "name" => "John Wick",
            "prodi" => "Sistem Informasi"
        ),
        array(
            "name" => "John Doe",
            "prodi" => "Teknik Informatika"
        )
    )
);

$data2_json = json_encode($data2);
var_dump($data2_json);
```

json_decode()

```
<?php
$str1 = '
{
    "name":"John Thor",
    "prodi":"Informatika",
    "email":"john.thor@umn.ac.id"
}
';

$str2 = '
{
    "students":[
        { "name":"John Thor", "prodi":"Informatika" },
        { "name":"John Wick", "prodi":"Sistem Informasi" },
        { "name":"John Doe", "prodi":"Teknik Informatika" }
    ]
}
';

$data1 = json_decode($str1, associative: true);
$data2 = json_decode($str2, associative: true);

echo "<pre>";
var_dump($data1);
var_dump($data2);
```


SUMMARY

After finishing this module, you should be able:

- to understand various predefined variables in PHP
- to understand various predefined functions in PHP
- to understand how Form submission works in PHP
- to understand HTTP Cookies
- to understand how to encode and decode json

NEXT WEEK'S OUTLINE

- PHP Object-Oriented Programming
 - Class & Object
 - Constructor
 - Exception Handling



next week

REFERENCES

- <https://www.w3schools.com/PHP/default.asp>
- <https://www.php.net/docs.php>
- Luke Welling & Laura Thomson. 2017. PHP and MySQL Web Development (5th ed)
- Robin Nixon. 2021. Learning PHp, MySQL & JavaScript.

Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.